

Mechanical and Electrical Properties of Novel Ultra-thin Layered Ferroelectric Bismuth Oxide Structures from First Principal Calculations

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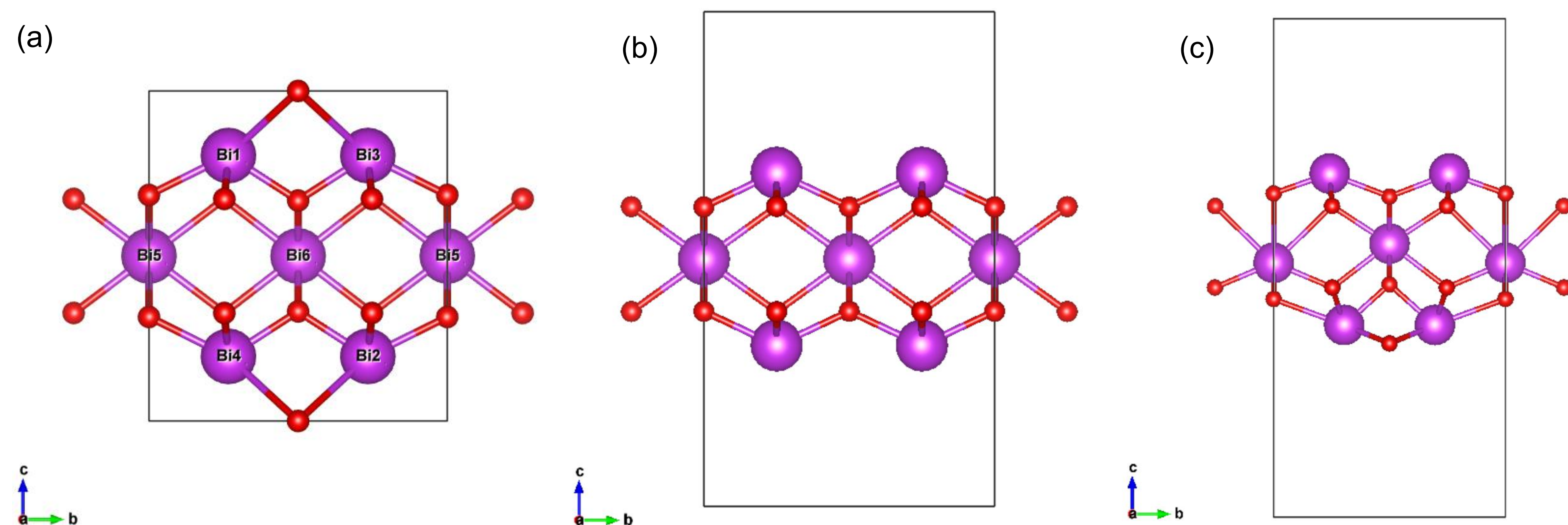
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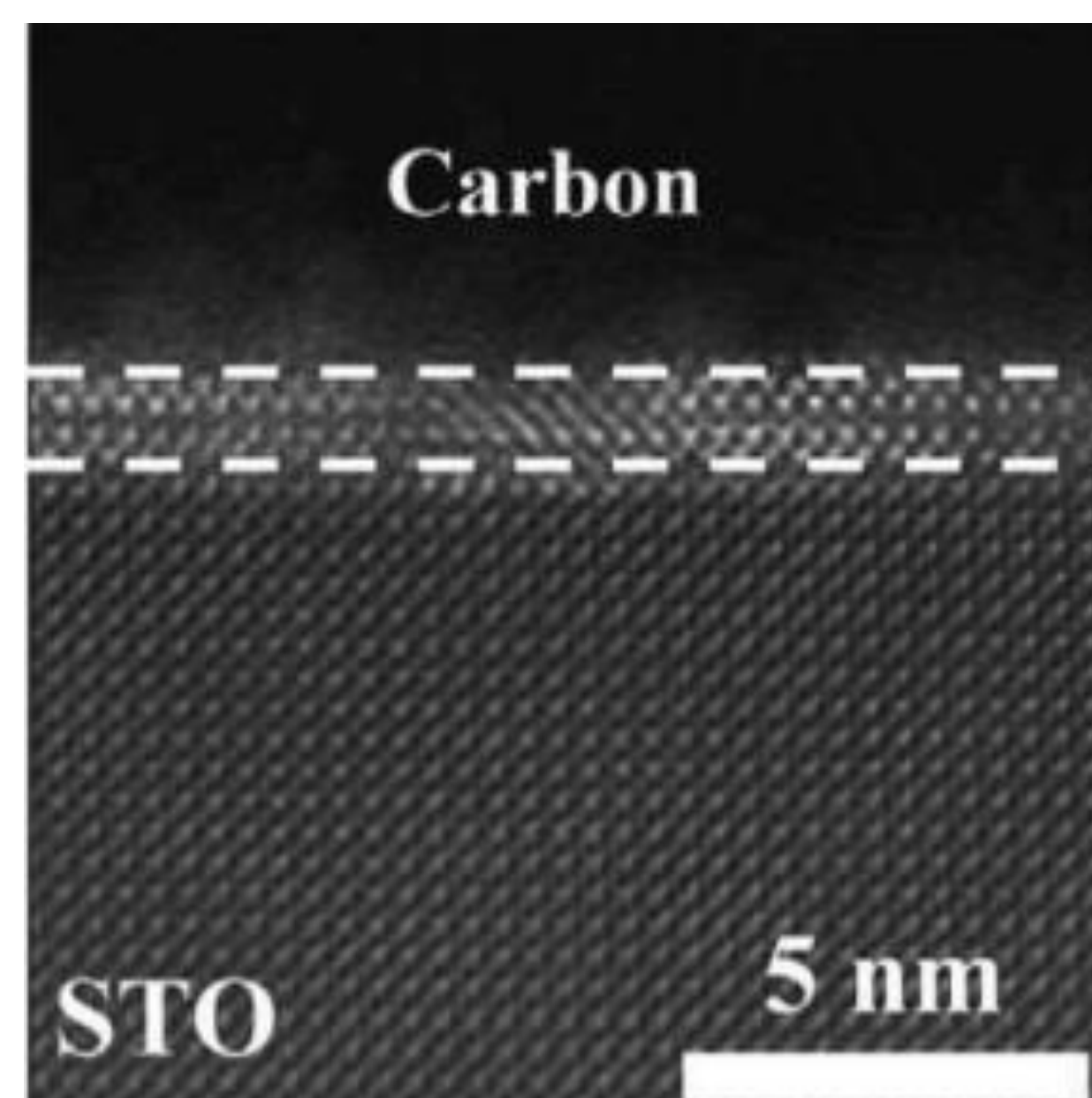
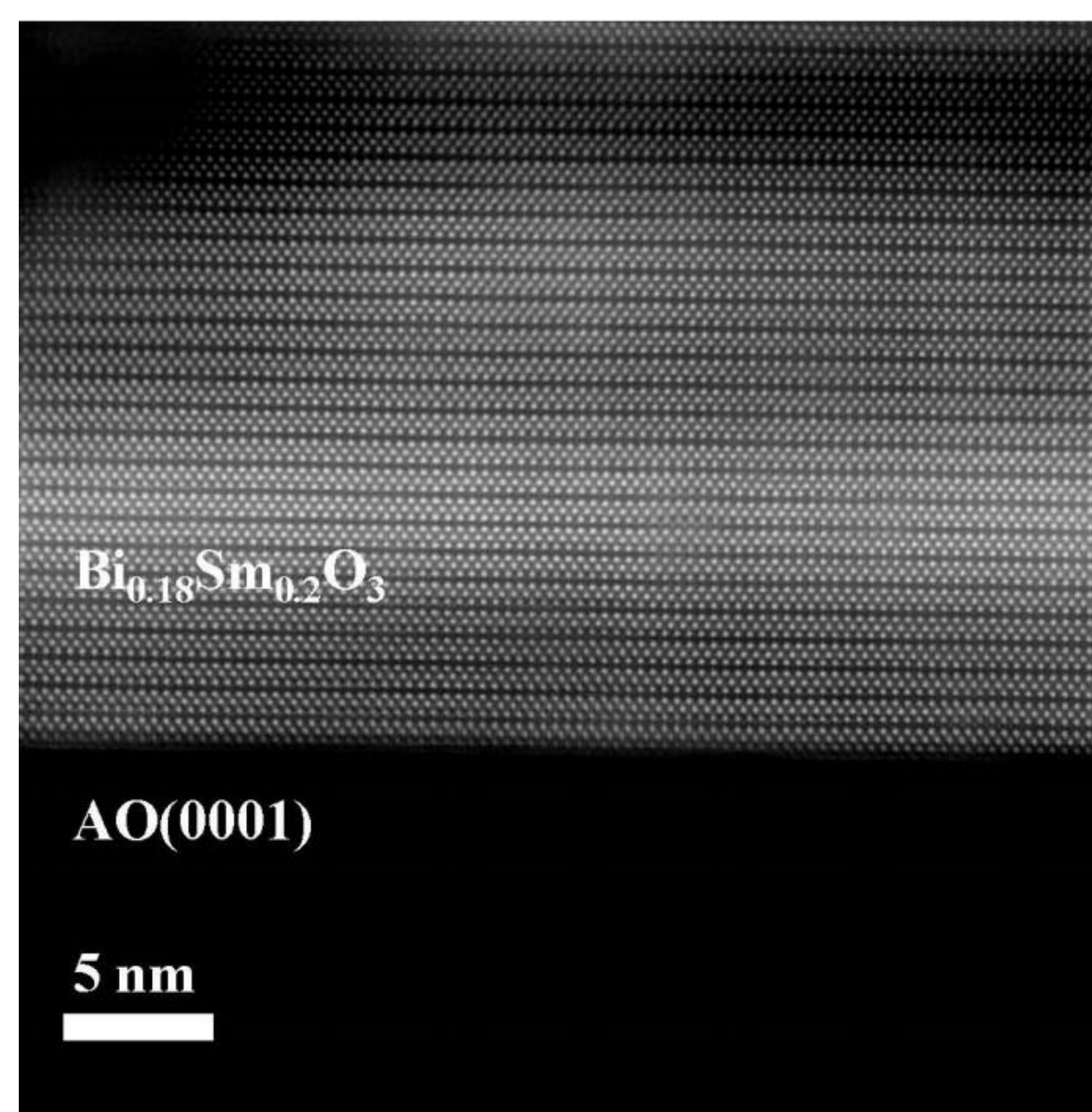
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Ferroelectric materials at the nanometre scale play a crucial role in the miniaturization of essential electronic components, such as memory devices, capacitors, piezoelectric sensors, and thin-film transistors. Recent research has proposed a new Bismuth Oxide structure that retains ferroelectricity down to the 1-nanometer scale. In our study, we applied first-principles calculations to both the bulk and thin-layered forms of this novel structure. Using density functional theory (DFT), we successfully confirmed the bulk paraelectric, bulk ferroelectric, and layered paraelectric phases. Phonon calculations allowed us to identify distortion patterns, and our band structure analysis revealed a band gap of 1.57 eV, further demonstrating the material's ferroelectric properties. This work offers additional insights into the potential applications of this material and similar ones in future research and industrial developments.

Structural relaxation



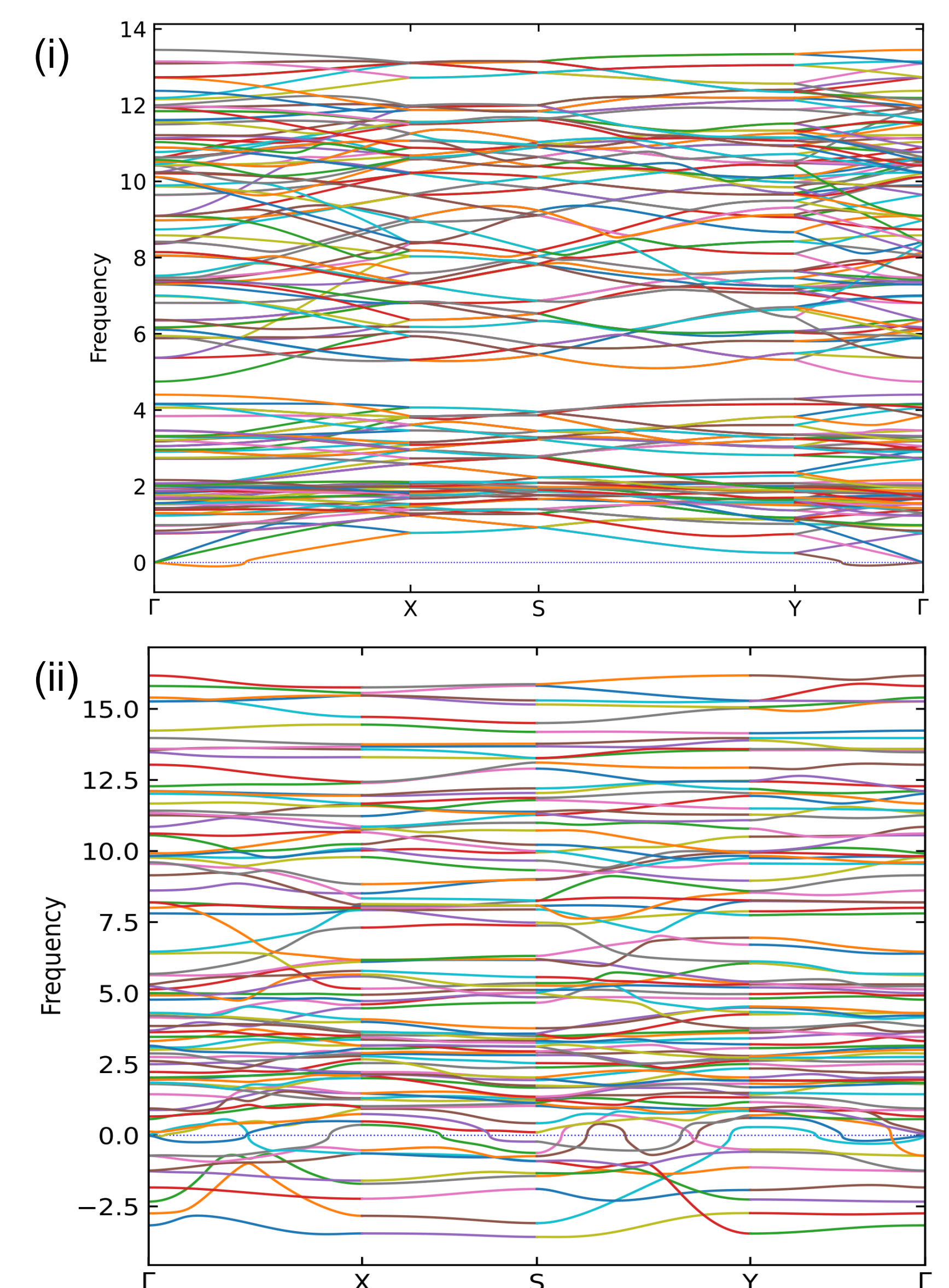
The relaxation analysis yielded the following results: (a) the bulk Bi_6O_9 paraelectric (PE) phase, exhibiting Pmmm group symmetry; (b) the monolayer Bi_6O_8 PE phase, also exhibiting Pmmm group symmetry; and (c) the monolayer Bi_6O_9 ferroelectric (FE) phase, characterized by Pmm2 group symmetry, which lacks inversion symmetry.



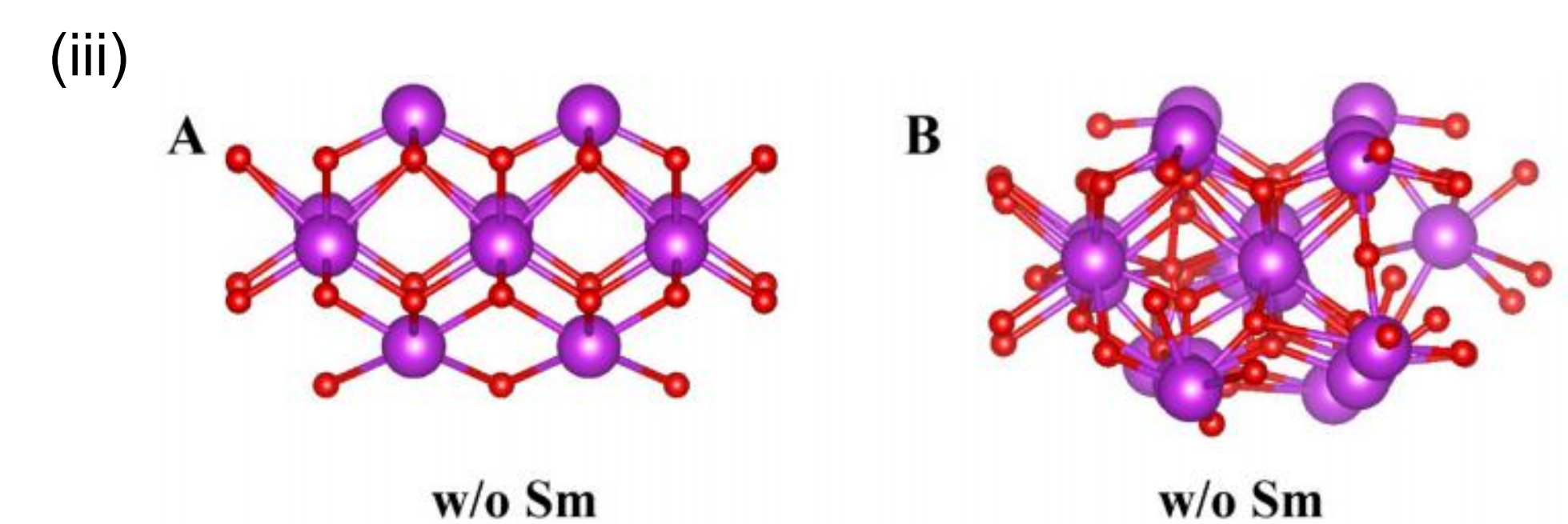
(Adapted from Figure 2D and Figure S10 in Yang et al., 2023)^[1]

The choice of the monolayer paraelectric phase was intentional. Experimental results on the left demonstrate the growth of a layered PE structure on Al_2O_3 (AO) and SrTiO_3 (STO)-based substrates^[1], with an oxygen atom shared between the monolayer and the substrate. Stability calculations confirmed that the actual layered paraelectric structure corresponds to Bi_6O_8 , without the additional oxygen layers.

Phonon bands

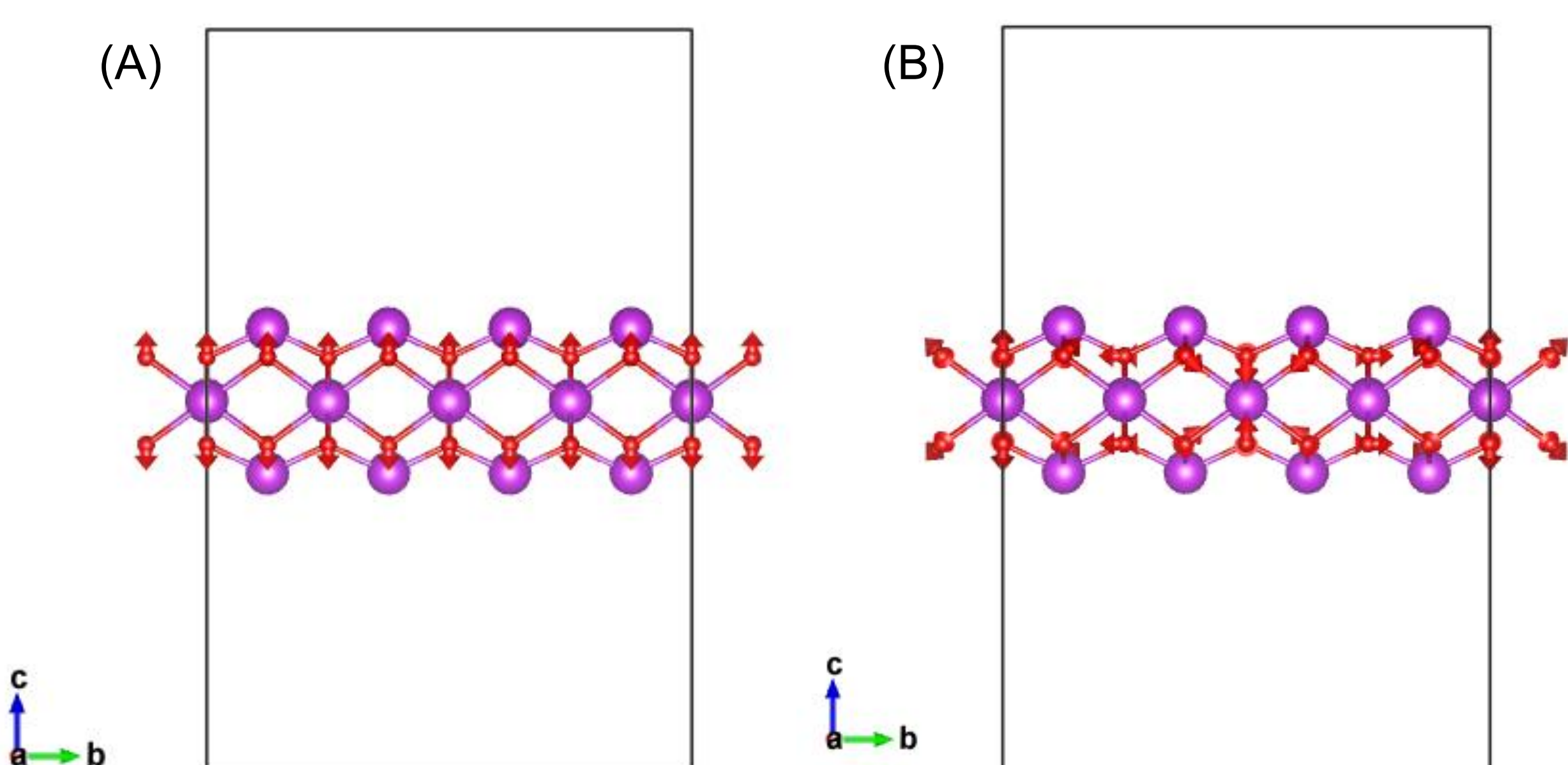


The latter shows a higher number of imaginary modes, indicating the structural instability of the monolayer FE structure. This observation aligns with experimental results, which suggest that the FE layered structure can only be stabilized through Sm-Bi substitution.



(iii) Instability of FE monolayer. (Adapted from Figure S8 in Yang et al., 2023)^[1]

visualization of phonon bands



Phonon modes offer key insights into distortion patterns and mechanical properties. Imaginary modes indicate instabilities at certain k-points, hinting at phase transitions. Additionally, these modes provide crucial information on the system's dynamics which may not be captured by traditional elastic-scattering techniques^[2]. Phonon mode visualization is a valuable tool as it provides a clear representation of how the atoms move collectively at specific points in the Brillouin zone.

The distortion of structures can be related to coupling of Landau levels resulting in certain ways of phase transitions.

The two graphs on the left visualize the real phonon modes of monolayer PE Bi_6O_8 from (A) $\Gamma(0,0,0)$ point and (B) $X(0.5,0,0)$ point in k-space. The arrows represent the atomic displacements (vibrations) corresponding to each atom.

[1] Qianqian Yang et al. „Ferroelectricity in layered bismuth oxide down to 1 nanometer. Science 379, 1218-1224 (2023). DOI:10.1126/science.abm5134

[2] M. E., and A. M. Glass, Principles and Applications of Ferroelectrics and Related Materials, Oxford Classic Texts in the Physical Sciences (online edn, Oxford Academic, 1 Feb. 2010)